

Homework Assignment Week 4
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DATA 620

Prompt:

Centrality measures can be used to predict (positive or negative) outcomes for a node.

Your task in this week's assignment is to identify an interesting set of network data that is available on the web (either through web scraping or web APIs) that could be used for analyzing and comparing centrality measures across nodes. As an additional constraint, there should be at least one categorical variable available for each node (such as "Male" or "Female"; "Republican", "Democrat," or "Undecided", etc.)

In addition to identifying your data source, you should create a high level plan that describes how you would load the data for analysis, and describe a hypothetical outcome that could be predicted from comparing degree centrality across categorical groups.

Assignment Begins Below:

- Data Set: Dune Characters (nodes) and their relationships to each other (edges)
- Categorical variable options: Character's Cultural Affiliation, character's House, Gender

I am choosing to look at dataset of important Dune characters (from the first and second books). I really enjoy this sci-fi universe, and as a very political universe, I thought that this would be interesting. Theoretically, a network here could show which characters are the most influential on the network of big political players. Or, it could show if a certain group is more influential or not. (Influential meaning high centrality.)

Plan:

- Locate the source data:
<https://www.kaggle.com/datasets/bac3917/frank-herberts-dune-characters>
- Load the data into my github repository to make it more easily accessible
- Load it into a Google Colab notebook
- Do some minor exploratory analysis and cleaning as is necessary
- Build the network
- Calculate centrality measures
- Hypothesize about what group may have the most influence in the Dune universe
- Test the hypothesis